University of Tokyo Graduate School Entrance Examination

2019 Master's Program — Physics Department

Part I (3 Hours)

Instructions

1. Write your answers to Problems I-1, I-2, and I-3 on their designated answer sheets.
2. You may use the back side of the answer sheet for Problem I-1.
3. Problem I-2 consists of two sub-problems (I-2A and I-2B). Problem I-3 consists of four independent sub-problems (I-3A, I-3B, I-3C, I-3D). Clearly indicate on the answer sheet (including the back side) which sub-problem each answer corresponds to.
4. Use the answer sheet horizontally. Write your examinee number and name at the top left of the front page (above the line). Do not write your examinee number or name elsewhere on the sheet. Do not write answers above or below the indicated answer fields.
5. Submit all three answer sheets (total of 3). The question booklet and draft paper will not be collected.
6. This question booklet consists of 7 pages including the cover.

Problem I-1 (Electromagnetism)

Consider the Maxwell equations in vacuum:

$$\nabla \cdot \vec{E} = 0 \quad \text{(A)}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{(B)}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{(C)}$$

$$\nabla \cdot \vec{B} = 0 \quad \text{(D)}$$

Let the electric and magnetic fields be $\vec{E}, \vec{B}$, and the current density $\vec{J}$. The constants are the permeability μ_0, permittivity ε_0, and the charge e and mass m of a particle. The velocity of the particle is $\vec{v}$. The equation of motion is:

$$\frac{d\vec{v}}{dt} = \frac{e}{m}\vec{E}$$

All quantities are in SI units.

1. Express $\vec{J}$ and $\partial \vec{J}/\partial t$ in terms of $\vec{E}$.
2. From (A) to (D), derive the wave equation for $\vec{E}$:

   $$\nabla^2 \vec{E} = \vec{E} + \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

   using $n, \mu_0, \varepsilon_0, e, m$ as necessary.
3. In the region $z > 0$, assume the dispersion relation:

   $$\omega^2 = c^2 k'^2 + \omega_p^2 \quad \text{where} \quad c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$$

   Express ω_p in terms of $n, \mu_0, \varepsilon_0, e, m$.

   In the region $z < 0$, a wave is incident from $-z$ direction:

   $$\vec{E}_i = \mathrm{Re}[E_0 e^{i(kz - \omega t)}]\hat{x} \quad \text{(with } E_0 \text{ real and along } x\text{)}$$

4. For $\omega > \omega_p$, determine the reflected and transmitted waves ($\vec{E}_r$, $\vec{E}_t$) at $z = 0$.
5. From (4), compute the time-averaged Poynting vector $\langle \vec{S} \rangle$ at $z = +0$.
6. For $\omega < \omega_p$, compute the field at $z = +0$.

Problem I-2A (Classical Mechanics)

A particle of mass M_s orbits a fixed mass M_p ($M_s \ll M_p$) under Newtonian gravity:

$$F = -G\frac{M_s M_p}{r^2}$$

Let (r, θ) be the polar coordinates. The radial and angular components of acceleration are given by:

$$a_r = \frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2, \quad a_\theta = \frac{1}{r}\frac{d}{dt}\left(r^2\frac{d\theta}{dt}\right)$$

1. Derive the expression for the areal velocity:

$$T = r^2\frac{d\theta}{dt}$$

2. Show that the following relations hold:

$$\frac{dr}{dt} = A\frac{d}{d\theta}\left(\frac{1}{r}\right), \quad \frac{d^2r}{dt^2} = B\frac{d^2}{d\theta^2}\left(\frac{1}{r}\right)$$

   and express the constants A and B in terms of T.

3. Using the elliptical orbit relation

$$r = \frac{a(1-\varepsilon^2)}{1+\varepsilon\cos\theta},$$

   show that the radial acceleration is

$$a_r = \frac{k_{sp}}{r^2}.$$

   Express k_{sp} in terms of a and T.

Based on the above, consider the approach Newton might have used. The planet is attracted by the Sun with force

$$F_p = a_r M_p = \frac{k_{sp}M_p}{r^2}.$$

According to Kepler's third law, k_{sp} is constant regardless of the planet.

On the other hand, assuming the Sun receives a force of the same form from the planet,

$$F_s = \frac{k_{ps}M_s}{r^2},$$

and since k_{ps} is not expected to depend on the Sun's properties, we let:

$$F_s = C,$$

where C is a constant.

From this reasoning, show that:

$$k_{sp} = -GM_s, \quad k_{ps} = -GM_p,$$

and hence derive the law of universal gravitation.

4. Provide a brief explanation that appropriately justifies the assumption made for C above.

Problem I-2B (Quantum Mechanics)

Consider a one-dimensional quantum harmonic oscillator with mass m and angular frequency ω. The normalized energy eigenstates $|n\rangle$ satisfy the commutation relation $[\hat{a}, \hat{a}^\dagger] = 1$, where $\hat{a}^\dagger$ and $\hat{a}$ are the creation and annihilation operators, respectively. The eigenstates are expressed as:

$$|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n|0\rangle, \quad (n = 0, 1, 2, 3, \dots)$$

Here, $|0\rangle$ is the ground state. Throughout this problem, $\hbar$ denotes the reduced Planck constant ($\hbar = h/2\pi$).

1. Given the position operator:

   $$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger),$$

   express the matrix elements $\langle n'|\hat{x}|n\rangle$ and $\langle n'|\hat{x}^2|n\rangle$ using Kronecker delta notation.

2. Suppose the oscillator is subject to perturbations represented by the following four potentials:

   $$V_1(x) = Ax, \quad V_2(x) = Bx^2, \quad V_3(x) = Cx^3, \quad V_4(x) = Dx^4.$$

   For each potential, determine whether a first-order energy shift arises for the state $|n\rangle$. Briefly explain your reasoning based on the symmetry properties of the Hamiltonian. Also, for each case where a shift occurs, express the magnitude of the first-order energy shift using the relevant variables among m, ω, n, A, B, C, D.

3. Find the first-order energy correction when both $V_1(x)$ and $V_2(x)$ are applied simultaneously.

Problem I-3A (Classical Mechanics)

A rocket of initial mass M_0 is launched vertically upward in a uniform gravitational field with gravitational acceleration g. The rocket expels fuel downward at a constant speed u relative to the rocket. As fuel is expelled, the mass of the rocket decreases.

Let $t = 0$ be the launch time. At any time t, let the mass of the rocket be $M(t)$. Derive the equation of motion for the rocket and find its velocity $v(t)$. Assume the direction upward is positive.

Problem I-3B (Statistical Estimation)

Let a discrete random variable k follow the Poisson distribution:

$$P_\lambda(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad (k = 0, 1, 2, \dots)$$

This distribution is called the Poisson distribution with parameter λ.

1. Derive the mean and variance of the Poisson distribution with parameter λ.
2. Consider a measurement in which particles emitted from a substance are detected. Assume that the number of detected particles follows a Poisson distribution. Determine how many particles must be detected in order for the standard deviation of the detected count to be less than 1% of its mean value.

Problem I-3C (Quantum Mechanics)

Consider a quantum system of two identical spin-1/2 particles bound in one dimension. The Hamiltonian is given by:

$$H = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2}\right) + \lambda |x_1 - x_2|$$

Here, x_1 and x_2 are the position coordinates of the two particles, m is their mass, $\hbar$ is the reduced Planck constant, and λ is a positive constant.

Determine the total spin of the system in the ground state and first excited state. Provide a brief explanation for your answer.

Problem I-3D (Mathematical Physics)

Let the vector quantity $\vec{r} = (x, y, z)$ follow a Gaussian distribution:

$$f(\vec{r}) = A \exp\bigl(-\beta|\vec{r}|^2\bigr)$$

Determine the constant A such that the distribution is normalized:

$$\int d^3\vec{r}\, f(\vec{r}) = 1$$

You may use the following integration formula:

$$\int_0^\infty s^2 e^{-as^2} ds = \frac{\sqrt{\pi}}{4a^{3/2}}$$

University of Tokyo Graduate School Entrance Examination

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Part II (3 Hours)

Instructions

1. Write your answers to Problems II-1, II-2, and II-3 on their designated answer sheets.
2. You may use the back side of the answer sheet for Problems II-1 and II-2.
3. Problem II-3 consists of two independent sub-problems (II-3A and II-3B). Clearly indicate on the answer sheet (including the back side) which sub-problem each answer corresponds to.
4. Use the answer sheet horizontally. Write your examinee number and name at the top left of the front page (above the line). Do not write your examinee number or name elsewhere on the sheet. Do not write answers above or below the indicated answer fields.
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Problem II-1 (Statistical Mechanics)

Consider a model in which each molecule consists of two atoms bound by a potential. Suppose that N such molecules are contained in a volume V, and the typical separation between atomic nuclei is much smaller than the intermolecular spacing $(V/N)^{1/3}$. Assume the interactions between molecules can be neglected. Let the Boltzmann constant be k_B.

1. Consider a model in which the distance between the two atomic nuclei in a molecule is strictly fixed at a value a. Within the framework of classical statistical mechanics, express the pressure $P(T,V,N)$ and the heat capacity at constant volume $C_V(T,V,N)$ in thermal equilibrium at temperature T. Briefly explain the underlying reasoning. A detailed calculation is not required.

2. Now consider a model in which the two nuclei in a molecule interact via a harmonic potential. That is, the interaction energy between two atoms at positions $\vec{r}_1$ and $\vec{r}_2$ is given by:
$$\frac{1}{2}k(|\vec{r}_1 - \vec{r}_2| - a)^2$$
Assuming $ka^2 \gg k_BT$, determine the pressure $P(T,V,N)$ and the heat capacity at constant volume $C_V(T,V,N)$ in thermal equilibrium at temperature T, using classical statistical mechanics. Briefly explain the method. Detailed calculations are not necessary.

3. Consider the setup in (2) using quantum statistical mechanics. Assume that atoms are identical bosons and follow Bose statistics. Let the Planck constant be $\hbar = h/2\pi$.

   For the potential in (2), suppose the vibrational energy levels are $T_0, T_1, T_2, \ldots$. Let the temperature be such that the molecules are primarily in the ground state and first excited state. Express the temperature T at which the number of molecules in the first excited state becomes twice that of those in the ground state. You may use $\hbar$ and k_B in your expression.

4. Name the remarkable physical phenomenon observed in the temperature regime $T \ll T_0$.

5. Assume $T_0 \ll T_1 \ll T_2$. For each of the three temperature regimes:

   (i) $T_0 \ll T \ll T_1$

   (ii) $T_1 \ll T \ll T_2$

   (iii) $T \gg T_2$

   write the heat capacity and briefly explain the reasoning.

6. At room temperature (around 300 K), the heat capacity of a two-atom gas molecule is approximately $\frac{5}{2}Nk_B$. In order to be consistent with this observation, the angular frequency of the harmonic oscillator $\omega = \sqrt{k/m}$ must be significantly greater than a certain value ω_c. Estimate this value ω_c.

   Use: $h = 10^{-34}\,\mathrm{m^2kg/s}$, $k_B = 10^{-23}\,\mathrm{m^2kgs^{-2}K^{-1}}$.

Problem II-2 (Quantum Mechanics)

Consider a particle with mass m and charge $e > 0$ confined to move on a ring of radius R on a horizontal plane. The ring lies in the xy-plane and is centered at the origin. A very thin solenoid runs vertically along the z-axis through the center of the ring, producing a magnetic flux Φ through the interior of the ring but no magnetic field along the ring itself. The particle is constrained to move only on the circumference of the ring. Let $\hbar = h/2\pi$.

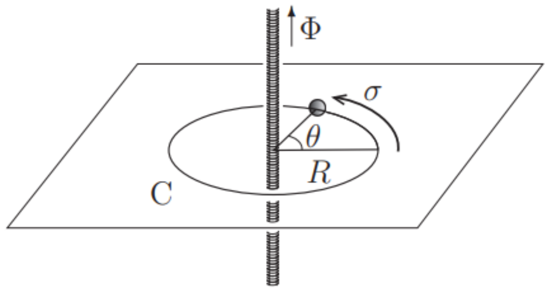

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Figure 1: A ring of radius R pierced by a magnetic flux Φ through its center.

1. When $\Phi = 0$, write the time-independent Schrödinger equation for the particle and determine the energy eigenvalues and normalized eigenfunctions.

2. For $\Phi \neq 0$, the tangential component of the vector potential contributes to the Hamiltonian. Express this contribution and modify the Schrödinger equation accordingly. Let e and $\hbar$ appear in your answer.

3. In general, let the canonical momentum be p_θ, and include the vector potential via minimal coupling $p_\theta \to p_\theta - eA$. Derive the energy eigenvalues in this case. Assume the magnetic flux Φ is fixed at a constant value.

4. Plot the energy spectrum as a function of Φ. Identify the flux values at which the ground state energy reaches a minimum. Express your answer in terms of the magnetic flux quantum $\Phi_0 = h/e$.

5. Suppose the magnetic flux is slowly varied from $\Phi = 0$ to Φ_0, and the system remains in its lowest energy state. Plot the probability density of the wave function at $\Phi = 0$ and $\Phi = \Phi_0$. If the ground state changes from $n = 0$ to $n = -1$ during this variation, describe where the peak of the probability density shifts.

Now consider the potential:

$$V(\sigma) = \lambda \cos\left(\frac{2\sigma}{R}\right),$$

where λ is a positive constant. Investigate how the energy spectrum changes when this potential is added, using properties of Mathieu functions.

The Mathieu function $\mathrm{me}_\nu(\xi, q)$ satisfies the differential equation:

$$\frac{d^2\varphi(\xi)}{d\xi^2} + (a - 2q\cos(2\xi))\varphi(\xi) = 0 \tag{A}$$

when q is fixed and a is treated as an eigenvalue parameter.

The Mathieu functions satisfy:

$$\mathrm{me}_\nu(\xi + \pi, q) = e^{i\nu\pi}\mathrm{me}_\nu(\xi, q), \quad \mathrm{me}_\nu(\xi, 0) = e^{i\nu\xi}$$

Here, $\nu = \nu(a, q)$ is a real number known as the characteristic exponent, determined by a and q. When ν is not an integer, the two functions me_ν and $\mathrm{me}_{-\nu}$ form a linearly independent pair of solutions to Eq. (A).

6. Show that the Schr"odinger equation for this system reduces to Eq. (A) by making the substitution $\xi = \sigma/R$, $\psi(\sigma) = e^{ik\sigma}\varphi(\xi)$ and choosing an appropriate k. Express a, q, Φ_0, λ, and the energy eigenvalue E in terms of the system parameters.

7. Let k, a, and q be those determined in (6). For the wavefunction $\psi(\sigma) = e^{ik\sigma}\mathrm{me}_\nu(\xi, q)$ to satisfy periodic boundary conditions, determine the values of ν that are allowed.

8. Figure 2 depicts the relation between a and ν for fixed q. In particular, when $\nu = 1.0$, there are two values of a near $a = 1.0$. This indicates that for certain q in the range $0 < q < 1$, there are values of a for which ν is not real. Determine the value of q for which this occurs. Then, estimate the minimum value of λ for which these phenomena appear using the q-λ relation found in (6). If the first and second excited state energy levels merge in Figure 2, explain that part.

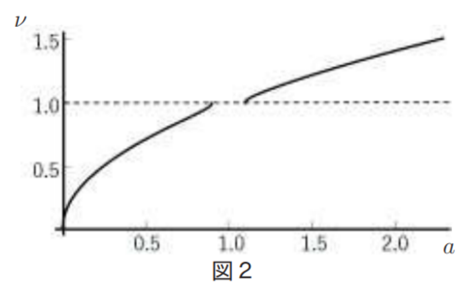

Figure 2: Figure 2

Problem II-3B (Mathematical Physics)

Let $T(x,t)$ be a temperature field defined on the one-dimensional interval $[0,L]$, satisfying the diffusion equation:

$$\frac{\partial T(x,t)}{\partial t} = D\frac{\partial^2 T(x,t)}{\partial x^2} \tag{B}$$

and the boundary conditions:

$$\left.\frac{\partial T(x,t)}{\partial x}\right|_{x=0} = 0, \qquad \left.\frac{\partial T(x,t)}{\partial x}\right|_{x=L} = 0 \tag{C}$$

1. For any continuous function $T(x,t)$ defined on $[0,L]$ with finite values, show that it can be expressed in terms of an appropriate set of real functions $\phi_n(x)$ $(n = 0, 1, 2, \dots)$ as:

   $$T(x,t) = \sum_{n=0}^{\infty} a_n(t)\phi_n(x)$$

   If $T(x,t)$ satisfies Eq. (B) and the boundary conditions (C), determine the corresponding form of the functions $\phi_n(x)$, and show that the equation for $a_n(t)$ involves constants m_n satisfying an ODE of the form:

   $$\frac{da_n(t)}{dt} = -m_n a_n(t)$$

   Specify the functions $\phi_n(x)$ and the constants m_n explicitly, and derive the ODE for $a_n(t)$.

2. Define $T_n(t)$ by:

   $$T_n(t) = \int_0^L dx\, \phi_n(x)T(x,t)$$

   If $a_n(0) \neq 0$, derive the expression for $T_n(t)$ and show that the result involves exponential decay.

3. Suppose that at $t = 0$, the temperature profile is given by:

   $$T(x,0) = A\left[1 - \left(\frac{2x}{L} - 1\right)^2 + \frac{1}{3}\left(\frac{2x}{L} - 1\right)^3\right]$$

   where A is a constant. Compute $\int_0^L dx\, T(x,0)$ for arbitrary L.

Problem II-3C (Experimental Physics)

Consider a method for measuring the momentum of a charged particle. As shown in Figure 1, a uniform magnetic field with magnitude B is applied perpendicular to the xy-plane (into the page) over a region D defined by $0 \le y \le L$. A positively charged particle enters vertically from $y = 0$ and moves through the region D under the influence of the magnetic field.

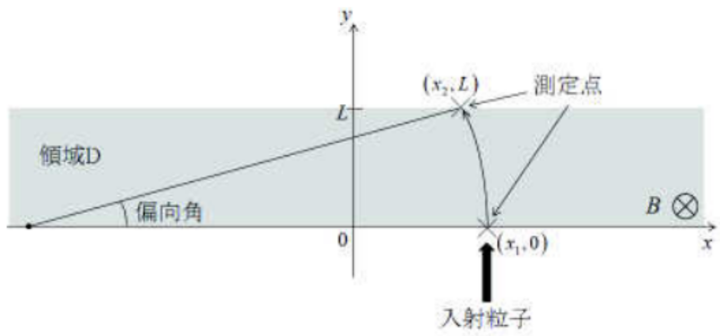

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Figure 3: Setup for measuring momentum using circular motion in a magnetic field.

1. Inside region D, the particle moves along a circular arc. Express the radius of this circular path as ρ, and express the particle's momentum p in terms of ρ.

2. Suppose the particle exits region D at $y = L$ and its position is measured. If the particle is observed at $y = 0$ with position $x = x_1$, and at $y = L$ with $x = x_2$, determine the radius ρ in terms of x_1 and x_2, assuming the deflection angle θ shown in Figure 1 is small and the momentum is sufficiently large.

3. From the measured positions x_1 and x_2, determine the uncertainty in the momentum p assuming the measurement uncertainties of x_1 and x_2 are σ_{x1} and σ_{x2}, respectively. Derive the expression for the standard uncertainty σ_p in p. Assume that the measurement errors are independent and the relative error is sufficiently small.

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Part III (3 Hours)

Instructions

1. Write your answers to Problems III-1, III-2, and III-3 on their designated answer sheets.
2. You may use the back side of the answer sheet for Problem III-1.
3. Problems III-2 and III-3 each consist of two independent sub-problems: III-2A, III-2B and III-3A, III-3B, respectively. Clearly indicate on the answer sheet (including the back side) which sub-problem each answer corresponds to.
4. Use the answer sheet horizontally. Write your examinee number and name at the top left of the front page (above the line). Do not write your examinee number or name elsewhere on the sheet. Do not write answers above or below the indicated answer fields.
5. Submit all three answer sheets (total of 3). The question booklet and draft paper will not be collected.
6. This question booklet consists of 9 pages including the cover.

Problem III-1 (Mechanics)

Consider the motion of a system of point masses connected by springs with natural length a and spring constant k.

We first consider a system of three point masses connected as shown in Figure 1. The point masses move only in the axial direction. The ends of the springs are fixed at positions a and $4a$, i.e., the total length is $L = 4a$. Let the positions of the three masses be $a + x_1$, $2a + x_2$, and $3a + x_3$, respectively. Here, x_1, x_2, x_3 represent the displacements from their equilibrium positions.

Let the central mass have mass M and the two side masses have equal mass m. Let $\omega_0 = \sqrt{2k/m}$.

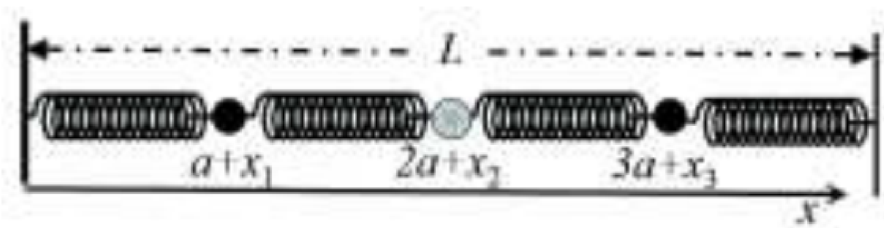

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Figure 4: Three-mass spring system.

1. Fix the positions of the two side masses such that $x_1 = 0$ and $x_3 = 0$. Find the displacement $x_2(t)$ of the central mass as a function of time. Assume initial conditions $x_2 = X$ and $dx_2/dt = V$ at $t = 0$.

2. Now release the constraints on the side masses so that all three masses can move. Find the normal mode frequencies of the system. There are three normal modes; denote the angular frequencies in ascending order as ω_1, ω_2, ω_3.

3. Write the equations of motion for the displacements $x_j(t)$ ($j = 1, 2, 3$), and assume the solutions take the form $x_j(t) = A_j \cos \omega t$. Let ω_1^2, ω_2^2, and ω_3^2 be expressed in terms of ω_0 and the mass ratio $\beta = m/M$. Let A_j be real constants.

4. Express the mode amplitudes for $\beta = 1$ using two of the amplitudes, such as A_2 and A_3, in terms of A_1.

5. Vary the mass M of the central mass and plot the dependence of ω_1^2/ω_0^2, ω_2^2/ω_0^2, and ω_3^2/ω_0^2 on $\beta = m/M$. Use horizontal axes from $0 \leq \beta \leq 2$. Use $\beta = 0$ and $\beta = 1$ to check your plot.

Problem III-2A (Lattice Vibrations)

Consider a one-dimensional chain of masses connected by identical springs, as shown in Figure 2. All masses except one have mass m, and one special mass at site $n = 0$ has mass M.

Let the equilibrium position of the n-th mass be at $x = na$, and its displacement be denoted x_n.

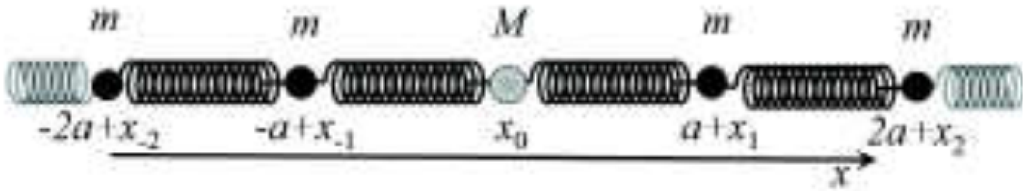

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Figure 5: One-dimensional chain with a single mass defect.

5. Suppose the M-mass at $n = 0$ is externally driven with angular frequency ω, such that:

$$x_n = \mathrm{Re}[B_n e^{i\omega t}], \quad B_n = B_0 b^{|n|},$$

where B_0 and b are complex constants. Substitute this into the equation of motion and express b in terms of ω and ω_0. Find the condition for which $|b| = 1$ and $|b| < 1$.

6. Now release the M-mass and allow it to move freely with the rest of the chain. A localized normal mode may appear under certain conditions. Use the same ansatz as in (5), and express the displacements as:

$$x_n = \mathrm{Re}[B_n e^{i\omega t}], \quad B_n = B_0 b^{|n|}$$

Select the values of $B_{\pm 1}$ such that the system remains symmetric and x_0 is the reference amplitude. Substitute the resulting expression into the equation of motion for x_0 and determine the condition under which a localized mode exists. Let $\beta = m/M$, and analyze the behavior of ω^2 as a function of β, especially the limiting behavior.

Problem III-2B (Electromagnetism)

Answer the following questions using SI units. Let the vacuum permittivity be ε_0.

1. Consider a uniformly charged solid sphere of radius a with total charge Q, centered at the origin. Using the distance r from the origin, determine the electric field produced by this sphere.

2. Determine the electrostatic potential energy U_0 of the uniformly charged sphere in (1).

   A π^- meson (charged pion) has the same charge as an electron but is more massive than a π^0 meson (neutral pion). It is observed that the energy of a charged π^- meson is higher than that of a neutral π^0. Assume this difference can be interpreted as a difference in electrostatic energy. Consider the following two models:

   Model I: A charged π^- meson is modeled as a uniformly charged sphere of radius a. A neutral π^0 meson is modeled as having no charge.

   Model II: A charged π^- meson is modeled as a pair of point charges $+2e/3$ and $-e/3$ separated by distance $2a$. A neutral π^0 meson is modeled as a superposition of two states: one with charges $+2e/3$ and $-2e/3$ separated by $2a$, and one with charges $+e/3$ and $-e/3$ also separated by $2a$. The electrostatic energy of the neutral meson is defined as the average of the energies of the two states.

3. Assuming the charged and neutral pions have the same electrostatic energy in Models I and II, find the ratio a/b, where b is the separation distance used in Model II.

Problem III-2C (Statistical Mechanics)

Consider a one-dimensional chain of N elements, where each element can independently take on a state $\sigma_i = +1$ or $\sigma_i = -1$. The energy and length contributions of each element are given by $\epsilon + \sigma_i \Delta$ and $l + \sigma_i a$, respectively. Let l, a, ϵ, Δ be constants.

If an external force f is applied to stretch the chain, the total energy of the system is:

$$E(\{\sigma_i\}) = \sum_{i=1}^{N} (\epsilon + \sigma_i \Delta) - fL \tag{A}$$

where the total length of the chain is:

$$L = \sum_{i=1}^{N} (l + \sigma_i a) \tag{B}$$

Let A be the physical quantity of interest, and assume the system is in a canonical distribution at temperature T. Use the Boltzmann constant k_B.

1. Find the probability that $\sigma_i = 1$ at temperature T.

2. Find the variance $\langle (\sigma_i - \langle \sigma_i \rangle)^2 \rangle$.

3. If the applied external force changes from f to $f + \delta f$ at constant temperature, and the expected length of the polymer changes from L to $L + \delta L$, show that for small δf, the change in length satisfies:

   $$\delta L \approx k_{\text{eff}} \delta f$$

   Find the effective spring constant k_{eff}.

4. Generalize Eq. (A) to the form:

   $$E(\{\sigma_i\}) = E_0(\{\sigma_i\}) - fL$$

   and express the fluctuation of the chain length $\langle (L - \langle L \rangle)^2 \rangle$ using Eq. (B). Assume that $E_0(\{\sigma_i\})$ does not depend on f.